[INCH-POUND]
A-A-3004
December 18, 1996

COMMERCIAL ITEM DESCRIPTION

PAD, FLOOR POLISHING MACHINE

1. SCOPE

1.1 Scope. This Commercial Item Description covers floor maintenance pads used for buffing, scrubbing, cleaning, stripping, burnishing, and polishing floors, suitable for mounting on power driven, rotary disk type floor machines.

2. CLASSIFICATION

2.1 Classification. The floor pads shall be of the following types, classes and sizes, as specified (see 7.1).

Type I - Nylon, 11/16 inch (17.5 mm) minimum thickness, for operation at up to 350 RPM.

Class 1 - Buffing (Tan)
Class 2 - Scrubbing (Green)
Class 3 - Stripping (Black)

Type II - Curled natural hair blended with synthetic fibers, 1 inch (25.4 mm) minimum thickness, for operation from 750 to 2500 RPM.

Class 1 - Buffing (Orange)
Class 2 - Scrubbing (Light Green)
Class 3 - Stripping (Dark Green)
Class 4 - Burnishing (Light Gray)

Type III - Polyester or polyester blended with nylon, 7/8 inch (22.2 mm) minimum thickness, for spray operation from 150 to 750 RPM.

Class 1 - Buffing (Red)
Class 2 - Cleaning (Blue)
Class 3 - Stripping (Brown)
Class 4 - Polishing (White)

The floor pads shall be of the following diameters when ordered by size (see 7.2):

Size 12 - 12 inches (305 mm)	Size 18 - 18 inches (457 mm)
Size 13 - 13 inches (330 mm)	Size 19 - 19 inches (483 mm)
Size 14 - 14 inches (356 mm)	Size 20 - 20 inches (508 mm)
Size 15 - 15 inches (381 mm)	Size 21 - 21 inches (533 mm)
Size 16 - 16 inches (406 mm)	Size 22 - 22 inches (559 mm)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Engineering Branch (7FXEE), 819 Taylor St., Fort Worth, TX 76102.

FSC 7910

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Size 17 - 17 inches (432 mm)

3. SALIENT CHARACTERISTICS

3.1 Material. The materials used in the manufacture of floor maintenance pads shall not support the growth of bacteria and must be of such denier, density, strength, quality, and workmanship to produce a cohesive pad, uniform in structure, and reversible with equal performance on both sides. The bonding adhesives shall be such that the pads shall not disintegrate in water, detergent solutions, oil, wax, or other material normally encountered in floor maintenance operations.

3.2 Design. The floor pads shall be suitably designed and fabricated for use with rotary disk electric floor machines at the rated speeds (see 2.1) and shall be suitable for use on various types of floors. The materials of the pads shall be formed or bonded into a solid disk, constituting a single piece unified mat. Pads of spirally wound ribbons of interlocking fibers shall not be permitted. The floor pads shall be resilient so as to contact the floor over their entire area while in use. The pads shall have a scored or removable center hole 3-3/8 ±1/8 inch (85.72 ±3.18 mm) in diameter. The center hole shall be concentric with the outside diameter within 5/16 inch (7.94 mm) total runout. The sizes of the pads shall be as specified within a tolerance of plus 3/8 inch (9.52 mm) or minus 1/4 inch (6.35 mm) on the diameter.

3.3 Thickness. Thickness measurement is performed using a height gauge with a readability of 0.01 inch (0.254 mm) mounted above a suitable level surface, and a leveling plate having a diameter of 90.0 ±1.0 mm and a weight of 100.0 ±1.0 g. Each pad to be tested shall be conditioned to ensure that any irregularity in thickness caused by shipping and storage is eliminated. Conditioning is achieved by placing the pad on a flat, level, hard surface and allowing it to remain for 48 hours at room temperature. After 48 hours, place the pad on the height gauge. Place the leveling plate on top of the pad. Wait one minute and measure the height of the pad. Take four readings 90 degrees apart. Average the four readings and record the thickness. Pad thickness shall fall within the range of values specified in Table I.

Table I

Type, Class	Color	Thickness in inches (thickness in mm given in parentheses)	Thickness loss, dry, percentage	Thickness loss, wet, percentage
I, 1	Tan	0.70 to 1.25 (18 to 32)	35	N/A
I, 2	Green	0.70 to 1.25 (18 to 32)	25	30
I, 3	Black	0.70 to 1.25 (18 to 32)	20	25
II, 1	Orange	0.80 to 1.25 (20 to 32)	35	N/A
II, 2	Light Green	0.80 to 1.25 (20 to 32)	25	30
II, 3	Dark Green	0.80 to 1.25 (20 to 32)	20	25
II, 4	Light Gray	0.80 to 1.25 (20 to 32)	40	N/A
III, 1	Red	0.70 to 1.25 (18 to 32)	35	N/A
III, 2	Blue	0.70 to 1.25 (18 to 32)	30	35
III, 3	Brown	0.70 to 1.25 (18 to 32)	25	30
III, 4	White	0.70 to 1.25 (18 to 32)	45	N/A

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3.3.1 Thickness under load. The same test specimens which are conditioned and tested in accordance with 3.3 are also used for this test. Add a 3 lb (1.36 kg) weight concentrically on the leveling plate. Repeat procedure used in 3.3. Calculate the percentage of dry loss in thickness of the pad. This percentage shall be in accordance with Table I. After noting the results, soak the class 2 and class 3 test pads for 5 minutes in hot tap water, minimum temperature 120°F (48.9°C). After soaking, shake out the excess water and repeat the tests specified in 3.3. Calculate the percentage of wet loss in thickness of the pad. This percentage shall be in accordance with Table I.

3.4 Abrasiveness. The abrasiveness of the pads shall be measured through Schiefer abrasion testing (also called Schiefer cut testing). The average Schiefer cut shall fall within the range of values specified in Table II.

3.4.1 Schiefer abrasion test. Calibrate and balance the machine according to the manufacturer's instructions. Set the countdown revolution counter to 5000 revolutions and set the revolution counter to zero. Weigh a 4-inch (102 mm) diameter, 1/8-inch (3.18 mm) thick acrylic tile, Rockwell M ball hardness 90 to 105, to the nearest 0.001 gram (W1) and record the weight. Center and secure the tile on the lower turntable using 1/8-inch (3.18 mm) thick double-faced foam tape equally spaced around the underside of the tile. Saturate a 4-inch (102 mm) diameter piece of test pad with water and center it against the upper turntable. Adjust the machine so that the upper turntable (including attached pad) will exert a 5 lb (22.2 N) load on the lower turntable. Adjust the water dispenser rate to 40 to 60 drops of water per minute. Lower the upper turntable so the pad contacts the tile and start the machine. After 5000 revolutions, shut off the machine, remove the pad and discard it. Remove the acrylic tile, remove the tape and rinse the tile with water. Wipe it dry and weigh it to the nearest 0.001 gram (W2), and record the weight. Record the Schiefer cut as the difference between W1 and W2. For each type and class, conduct the test three times, calculating the average Schiefer cut.

Table II

Type, Class	Color	Schiefer cut (grams)
I, 1	Tan	0.00 to 0.25
I, 2	Green	0.40 to 3.06
I, 3	Black	1.36 to 5.96
II, 1	Orange	0.00 to 0.25
II, 2	Light Green	0.00 to 1.00
II, 3	Dark Green	0.32 to 2.00
II, 4	Light Gray	0.00 to 0.25
III, 1	Red	0.00 to 0.50
III, 2	Blue	0.98 to 2.62
III, 3	Brown	1.65 to 4.48
III, 4	White	0.00 to 0.25

3.5 Abrasive uniformity. The pad shall be thoroughly and uniformly impregnated with abrasive particles. The uniform distribution of abrasive particles in the cross section shall be within ±15%. The flat surface of the pad shall also show no concentrations of particles visible to the eye when viewed toward a bright light source.

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3.5.1 Abrasive uniformity test. Four 1-inch (25.4 mm) square sections shall be cut one from each quadrant of the sample pad. Each square cut from the pad shall be sliced into three approximately equal layers. The layers shall be identified as top, middle, and bottom layers. The four top layers shall be placed in the porcelain dish after treating, weighed to the nearest centigram, ashed (see 3.7.2) and, when cooled, weighed again. Subtract the vessel weight and record the weight of the abrasive content (residue). Repeat the procedure for the middle layers and the bottom layers. The maximum difference in weight of the abrasives for the three layers shall be ±15%.

3.5.2 Ashing procedure. Heat a suitably clean ashing vessel, such as a porcelain dish, approximately 85 to 100 mm in diameter, at 600°C for 15 minutes. Cool in a desiccator and weigh to the nearest centigram. Add samples of the floor pad to be analyzed, favoring pieces approximately 1/2 g each, confining each within the perimeter of the vessel. (This will help to prevent certain types of pad samples from floating out of the vessel). Weigh the vessel and contents. Compute and record the sample weight with the appropriate identification data. Drive off obnoxious organic volatiles under a fume hood until a dry char is obtained. Finish ignition of the ash in the muffle furnace. A furnace with regulated air ventilation is preferable because air circulation aids the ashing which may be completed in approximately 15 minutes. Without air circulation, one-half to one hour may be required. When ashing is complete, remove the vessel with ash to a refractory plate to cool, finishing the final cooling in a desiccator. Be careful to avoid expulsion of fluffy ash particles. When cooled to room temperature, weigh the vessel with ash.

3.6 Visual and dimensional examination. Remove and label four pads of the same type, class, and size for examination, each from a different carton. Each whole pad shall be examined for construction, workmanship, and conformance with dimensional requirements.

3.7 Test Equipment. The following test equipment is required for Schiefer cut and abrasive uniformity testing:

Gram scale calibrated in centigrams;
Ashing dish, 85 to 100 mm-diameter;
Gas burner for igniting specimens (Meeker type or equivalent);
Schiefer abrasion testing machine;
4-inch (102 mm) diameter, 1/8-inch (3.18 mm) thick acrylic tiles, Rockwell M ball hardness 90 to 105;
Other items as specified in the test paragraphs.

3.8 Workmanship. The floor pads shall be free from any defects, imperfections, or material that would adversely affect performance, serviceability, or appearance, or cause any undesirable effects during use. The fibers shall not become matted nor lose their resilience.

3.9 Measurement system. The values stated in inch-pound units are to be regarded as the standard. The metric values stated in parentheses are for informational purposes only. Testing equipment may be specified in metric units without mention of equivalent inch-pound units

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3.10 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4. REGULATORY REQUIREMENTS

4.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

6. PACKAGING

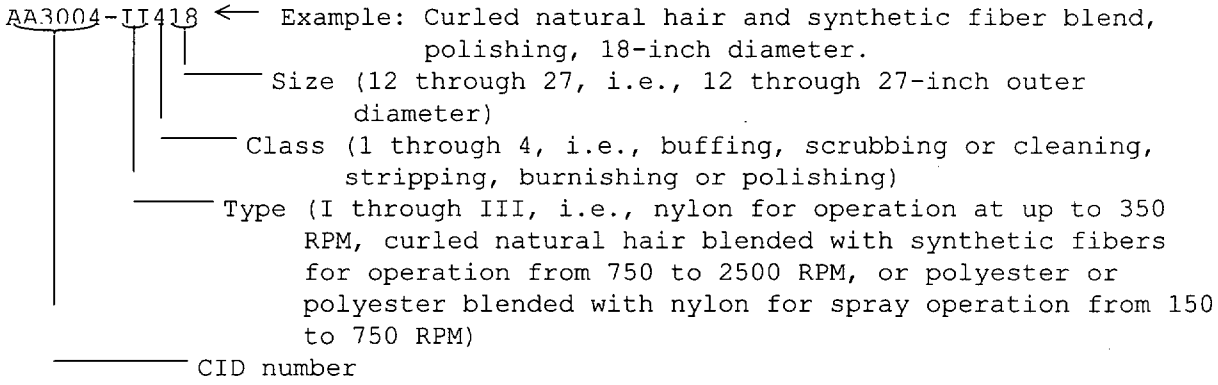
6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents (if applicable).

a. Title, number, and date of this Commercial Item Description.
b. Type, class, and size (see 2.1).
c. Preservation, packing, and marking desired.

7.2 Part Identification Number. The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



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7.3 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	Type	Class	Size
7910-00-820-7991	I	1	14
7910-00-820-7989	I	1	15
7910-00-820-7990	I	1	16
7910-00-820-9926	I	1	17
7910-00-820-9925	I	1	18
7910-00-820-9924	I	1	19
7910-00-820-9898	I	1	20
7910-00-820-7997	I	1	21
7910-00-820-7996	I	1	22
7910-00-820-9903	I	2	14
7910-00-820-9904	I	2	15
7910-00-820-9905	I	2	16
7910-00-820-9900	I	2	17
7910-00-820-9901	I	2	18
7910-00-820-9899	I	2	20
7910-00-820-9922	I	2	22
7910-00-820-9918	I	3	14
7910-00-820-9917	I	3	15
7910-00-820-9916	I	3	16
7910-00-820-9915	I	3	17
7910-00-820-9914	I	3	18
7910-00-820-9913	I	3	19
7910-00-820-9912	I	3	20
7910-00-820-9911	I	3	21
7910-00-820-9910	I	3	22
7910-00-985-6800	II	1	14
7910-00-985-6851	II	1	15
7910-00-985-6852	II	1	16
7910-00-985-6853	II	1	17
7910-00-985-6855	II	1	18
7910-00-985-6856	II	1	19
7910-00-985-6857	II	1	20
7910-00-985-6858	II	1	21
7910-00-985-6859	II	1	22
7910-00-985-6860	II	2	14
7910-00-985-6861	II	2	15
7910-00-985-6862	II	2	16
7910-00-985-6863	II	2	17
7910-00-985-6864	II	2	18
7910-00-985-6866	II	2	20
7910-00-985-6868	II	2	22
7910-00-985-6869	II	3	14
7910-00-985-6870	II	3	15
7910-00-985-6871	II	3	16
7910-00-985-6872	II	3	17
7910-00-985-6873	II	3	18
7910-00-985-6874	II	3	19
7910-00-985-6875	II	3	20
7910-00-985-6876	II	3	21

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7910-01-094-8423	III	1	14
7910-01-090-9827	III	1	16
7910-01-090-9828	III	1	18
7910-01-091-8960	III	1	20
7910-01-091-8961	III	1	22
7910-01-092-8504	III	2	14
7910-01-094-8424	III	2	16
7910-01-094-0032	III	2	18
7910-01-090-9829	III	2	20
7910-01-091-8962	III	2	22
7910-01-090-9830	III	3	14
7910-01-092-8505	III	3	16
7910-01-092-8502	III	3	18
7910-01-095-7831	III	3	20
7910-01-094-0033	III	3	22
7910-01-091-8957	III	4	14
7910-01-091-8958	III	4	16
7910-01-091-8959	III	4	18
7910-01-092-8503	III	4	20
7910-01-089-9084	III	4	22

MILITARY INTERESTS:

Preparing Activity:
GSA-FSS

Custodians
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Review Activities
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